Teacher attrition amid geographical, societal, and economic realities: Fifteen years of the Kansas teacher workforce

Chanh Bao Lam

College of Education
Kansas State University

April 3, 2026

EDCATS
Educators Powered by Purpose

1. Background
2. Research Questions
3. Data and Methods
4. Results
   - Contexts in the Kansas teacher workforce
   - RQ1. The extent to which Kansas teachers turned turnover
   - RQ2. Turnover among teacher, school, and student contexts
   - RQ3. Turnover and salary differential associations by teacher type
5. Implications

1 Background

2 Research Questions

3 Data and Methods

4 Results

5 Implications

Retaining teachers for student learning and school stability

- Teachers are the key resource to providing high quality and equitable educational opportunities to students (Darling-Hammond, 2010).
- Teacher staffing challenges have been chronic issues across the United States, significantly contributed by persistently high degrees of teacher attrition (García & Weiss, 2020; Goldhaber & Theobald, 2023; Nguyen et al., 2024).
- Teacher attrition disrupts student learning, diminishes school institutional knowledge, relational trust, collegial morality; and, perpetuates the feedback loop of staff replacement that impairs school stability (Hanushek et al., 2016; Henry & Redding, 2020; Ronfeldt et al., 2013; Sorensen & Ladd, 2020).

The need for understanding teacher attrition in Kansas

- Kansas is facing chronic teacher staffing challenges (Kansas State Department of Education, 2016, 2019, 2022, 2024), yet Kansas teachers and their mobility have not been substantially studied in the existing literature (Nguyen, 2020).
- Kansas's rural and urban dynamics, conservative influences, and industrial roots create a unique context to which evidence at the national level or from other states do not fully apply (Garay-Huamán & Irazábal-Zurita, 2021; Nguyen, 2020).
- Understanding teacher attrition specifically in Kansas is imperative to:
  - informing the state's policy considerations aimed at sustaining teachers, and
  - contributing to the collective understanding of teacher mobility in the U.S.

1 Background

2 Research Questions

3 Data and Methods

4 Results

5 Implications

Research Questions

RQ1. To what extent did Kansas public-school teachers turn over from 2010-2011 through 2023-2024?

This is critical for informing policymakers and school leaders on the current staffing challenges.

RQ2. To what extent did Kansas public-school teacher attrition differ by teacher type, including STEM, CTE, and SPED/ESOL,[1] school locale, and student racial and socioeconomic compositions?

This addresses variations in turnover across contexts, thereby informing targeted policy interventions. While SPED/ESOL teacher turnover is distinctive, understanding STEM and CTE attrition is important in the Kansas labor market landscape

RQ3. To what extent are teacher type (i.e., discipline) and salary predictive of turnover among Kansas public-school teachers?

This addresses ongoing debates on differential compensation among teachers, particularly those in hard-to-staff schools and subject areas, including discussions related to Missouri S.B. 727.

[1] Though there are distinctive differences between SPED and ESOL teachers, they could not be consistently identified in the data.

1 Background

2 Research Questions

3 Data and Methods

4 Results

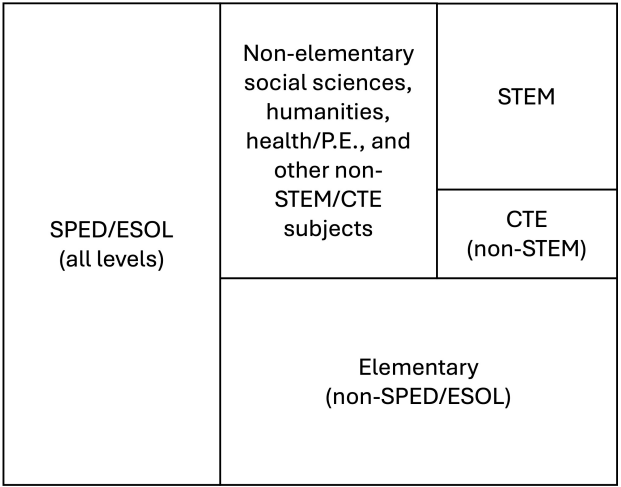
5 Implications

Data sources

- KSDE elementary and secondary school teacher data, 2010 through 2024:
  - Teacher age, race, gender, licensure areas, years of teaching, and degrees,
  - Teaching assignments, including course titles, grade levels, and FTE status,
  - Total salary: Base salary plus fringe benefits.
- National Center for Education Statistics's Common Core Data:
  - School size, grade levels, and school locale,
  - Student races, genders, and Free or Reduced-price Lunch (FRPL) eligibility.
- Nelson & Nguyen's (2023) Community Assets and Relative Rurality (CARR) data
  - Local services and amenities availability,
  - Local services and amenities accessibility, [2]
  - Local population density and land coverage.

[2] Examples: childcare centers, banks, schools, restaurants, libraries, arts centers, roads, and natural resource extractions.

Teacher types by teaching discipline



Analytic approaches

I employ descriptive and regression analyses. The base model is given as follows:

$$y_{isdct} = \beta_0 + \text{Ttype}'_{isdct}\beta_1 + \text{Work}'_{sdct}\beta_2 + \text{adjsal}_{isdct}\beta_3 + \text{TScov}'_{isdct}\beta_4 + \text{rel_urban}_d\beta_5 + \text{Flag}'_{isdct}\beta_6 + c + t + \epsilon_{isdct}. \quad (1)$$

This equation includes

- y_{isdct}: turnover outcome for teacher i from school s, district d, county c, in year t. y_{isdct} equals 1 if the teacher turned over, and 0 if they stayed;
- Ttype'_{isdct}: vector of teacher types, including STEM, CTE, other non-elementary (e.g., English, social studies, fine and performing arts, health and physical education, etc.), and SPED/ESOL;
- Work'_{sdct}: vector of school locales: urban, suburban, and high-minoritized and high-impoverished school indicators;
- adjsal_{isdct}: annual total salary (base salary plus benefits) in 2023-2024 dollars; and,
- Control factors including other teacher and school covariates, TScov'_{isdct}, the school community's resource availability and accessibility derived from CARR data, rel_urban_d, flags of abnormal records (e.g., FTE higher than 1.0), Flag'_{isdct}, and county- and year-fixed effects.

1 Background

2 Research Questions

3 Data and Methods

4 Results
- Contexts in the Kansas teacher workforce
- RQ1. The extent to which Kansas teachers turned turnover
- RQ2. Turnover among teacher, school, and student contexts
- RQ3. Turnover and salary differential associations by teacher type

5 Implications

1 Background

2 Research Questions

3 Data and Methods

4 Results

Contexts in the Kansas teacher workforce

RQ1. The extent to which Kansas teachers turned turnover

RQ2. Turnover among teacher, school, and student contexts

RQ3. Turnover and salary differential associations by teacher type

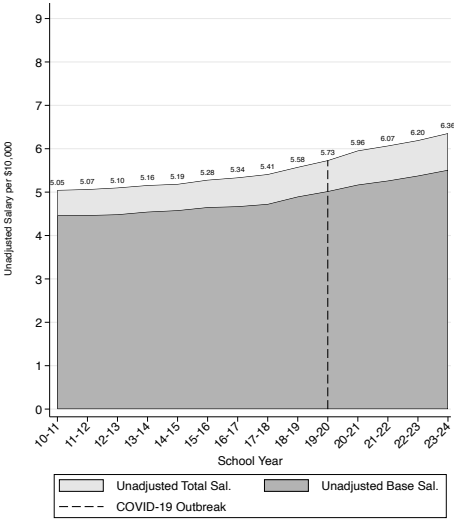
5 Implications

Kansas teacher characteristics and working conditions

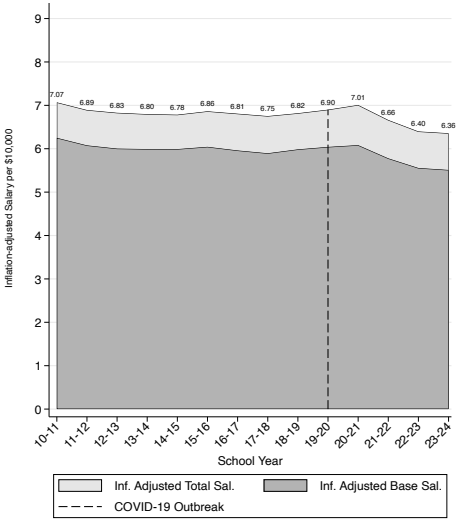
Teacher characteristics	Pooled average
Women (%)	76.2
Teacher of color (%)	6.2
Elementary teacher (%)	43.3
STEM teacher (%)	15.3
CTE teacher (%)	3.9
Other secondary (%)	28.3
SPED/ESOL teacher (%)	9.2
Observations	457, 346

Working conditions	Pooled average
City school (%)	26.1
Suburban school (%)	15.7
Rural/town school (%)	58.2
Percent students of color	34.4
Percent FRPL eligibility	49.2
Inf-adj total salary ($1,000)	67.8
Inf-adj base salary ($1,000)	59.4
Observations	457, 346

The stagnation of teacher salaries



Note: Line labels indicate unadjusted total salary.



Note: Line labels indicate inflation-adjusted total salary.

1 Background

2 Research Questions

3 Data and Methods

4 Results
- Contexts in the Kansas teacher workforce
- RQ1. The extent to which Kansas teachers turned turnover
- RQ2. Turnover among teacher, school, and student contexts
- RQ3. Turnover and salary differential associations by teacher type

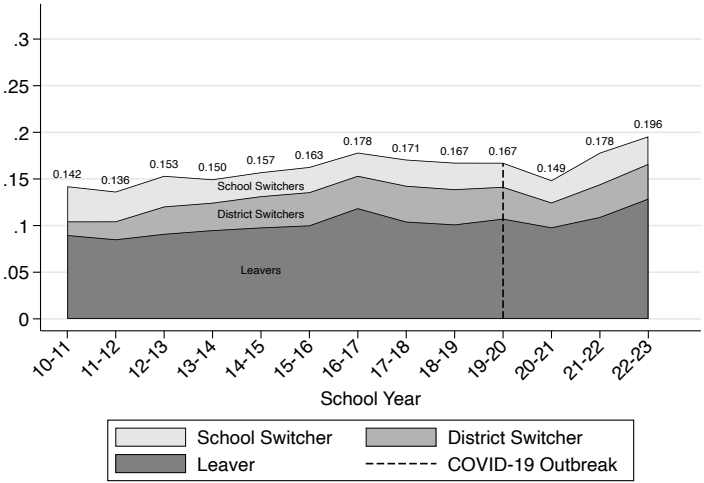
5 Implications

Overall turnover in the Kansas teacher workforce

Pooled turnover rates, 2010-11 through 2022-23

	Pooled average
Turnover (%)	16.2
School switcher	2.9
District switcher	3.1
Leaver	10.2
Observations	457,346

Teacher attrition over time



Note: Line labels indicate annual overall turnover rates (i.e., all switchers + leavers).

First takeaways

- Four in 25 Kansas teachers resign every year, comparable to the national average (16%, NCES, 2023), but more often exiting schools than switching.
- Turnover has sharply increased since COVID-19, reaching almost 20% in 2022-23, stressing Kansas's teacher staffing challenges ($1,810$ vacancies, KSDE, 2024).
- Stagnant teacher salaries and stagnant teacher diversity relative to student diversity are important considerations for sustaining the Kansas teacher workforce.

This connects us to RQ2

1 Background

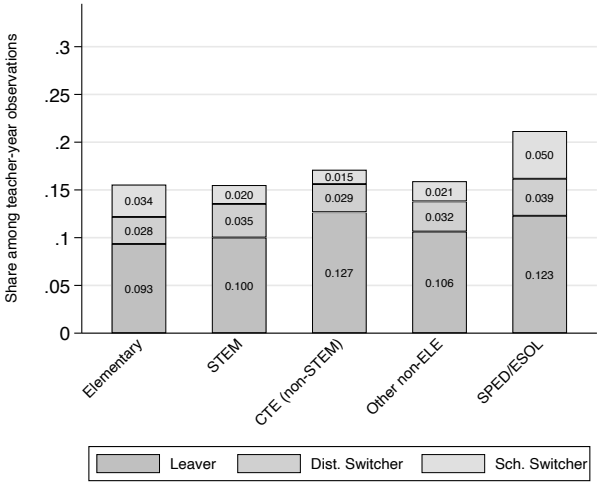
2 Research Questions

3 Data and Methods

4 Results
- Contexts in the Kansas teacher workforce
- RQ1. The extent to which Kansas teachers turned turnover
- RQ2. Turnover among teacher, school, and student contexts
- RQ3. Turnover and salary differential associations by teacher type

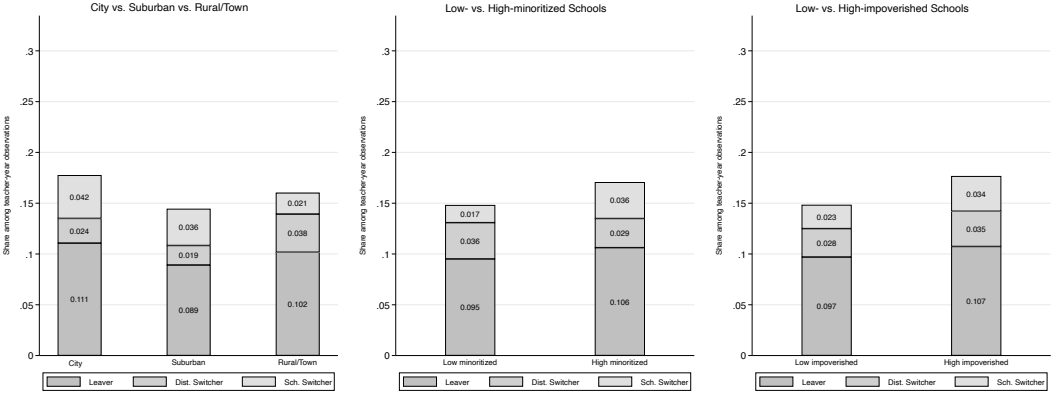
5 Implications

Turnover among teaching categories



Note: Inside-bar labels indicate form-specific pooled turnover rates.

Turnover across school locales and student body characteristics



Turnover regression by teacher, school, and student factors

	Overall turnover	School switcher	District switcher	Leaver
Elementary	−0.0109**	0.0055*	−0.0095**	−0.0089**
STEM	0.0094**	0.0033**	0.0044**	0.0047*
CTE (non-STEM)	−0.0033	−0.0010	−0.0029	−0.0013
SPED/ESOL	0.0331**	0.0186**	0.0024	0.0215**
City School	0.0095	0.0035	−0.0037^{+}	0.0110**
Suburban School	−0.0065	−0.0003	−0.0024	−0.0046
High Minoritized	0.0103**	0.0074**	0.0007	0.0042
High Impoverished	0.0177**	0.0103**	0.0049**	0.0071**
Inf-adj Total Salary per $10K	−0.0118**	−0.0031**	−0.0081**	−0.0044**
Rel. Urbanicity (percentile)	0.0004*	0.0007**	−0.0002**	0.0001
Observations	425210	368474	369541	399661

Note: $^{+}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$. County- and year-fixed effects are employed. Standard errors are clustered at the district level. Teacher type reference: other non-elementary, general education teachers (e.g., humanities, social studies, arts, physical education, etc.).

Second takeaways

Kansas teacher turnover varies across teacher, school, and student contexts.

- STEM teachers and teachers employed in city schools are more likely to leave public schools entirely.

  This reflects the dynamics of the Kansas context and the status of the teaching profession.

- Teachers employed in schools with relatively high percent of marginalized students tend to switch to another school in the district. Schools serving relatively high percent of impoverished students face the most difficult retention challenges.

  These patterns suggest potential disparities in teacher working conditions even among schools within the same district.

- Salary and benefits remain a significant factor of Kansas teacher attrition.

  This connects us to RQ3.

1 Background

2 Research Questions

3 Data and Methods

4 Results
- Contexts in the Kansas teacher workforce
- RQ1. The extent to which Kansas teachers turned turnover
- RQ2. Turnover among teacher, school, and student contexts
- RQ3. Turnover and salary differential associations by teacher type

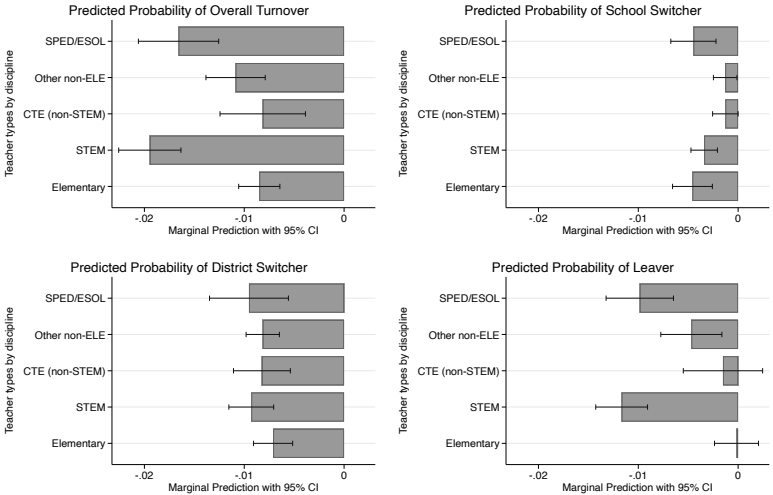
5 Implications

Turnover regression with teacher category and salary interactions

	Overall turnover	School switcher	District switcher	Leaver
Salary	-0.0109^{**}	-0.0013^{*}	-0.0081^{**}	-0.0047^{**}
Salary × Elementary	0.0024^{+}	-0.0033^{**}	0.0010	0.0045^{**}
Salary × STEM	-0.0086^{**}	-0.0021^{**}	-0.0011	-0.0070^{**}
Salary × CTE (non-STEM)	0.0027	0.0000	-0.0001	0.0032^{+}
Salary × SPED/ESOL	-0.0057^{**}	-0.0032^{**}	-0.0014	-0.0052^{**}
Observations	425,210	368,474	369,541	399,661

Note: $^{+}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$. County- and year-fixed effects are employed. Standard errors are clustered at the district level. Teacher type reference: other non-elementary, general education teachers (e.g., humanities, social studies, arts, physical education, etc.).

Marginal associations between turnover and salary by teacher type



Note: Full regression results are provided in Appendix C. County and Year Fixed Effects are employed.

Third takeaways

- Salaries predict retention, but to different extents across teaching disciplines.
- STEM and SPED/ESOL teachers are most likely retained for increases in compensation (salary & benefits) compared to teachers of other disciplines.
- The differential associations are most pronounced for leavers.

1 Background

2 Research Questions

3 Data and Methods

4 Results

5 Implications

Teacher characteristics and working conditions

- Kansas teachers remain less diverse relative to students. There have also been declines in certifications in high-turnover fields such as STEM and SPED/ESOL (results not shown). Strengthening and expanding pathways into teaching while diversifying the workforce remain key areas for consideration.
- Kansas's existing teacher recruitment programs (e.g., Computer Science Educator, Education Opportunity, Teacher Service scholarships, Registered Teacher Apprenticeship Model) can leverage this study's findings to further address workforce challenges in hard-to-staff/retain subject areas and locations.
- Turnover remains higher in underresourced schools and those serving higher percentages of marginalized students, including through within-district switching. This emphasizes the need for equitable resource allocation and greater attention to day-to-day teacher working conditions across school contexts.

Teacher pay and beyond

- Salary remains a significant factor, and salary increases are differentially associated with teacher retention across disciplines. Addressing teacher compensation is increasingly urgent given recent salary stagnation. Potential approaches may include targeted pay raises, prioritizing high-need areas while strengthening compensation for all teachers over the long run.
- Salary matters but has never been the sole concern of the educator workforce or public education. Evidence from teacher strikes shows concerns beyond pay to improved school resources and even labor rights, while districts with strikes do not necessarily have lower salaries than those without (Lyon et al., 2024).
- Supporting teacher retention, and broadly, sustaining school systems, requires attention to teacher characteristics, school conditions, and factors beyond the school such as pay policy or community context, as well as how these factors interact in relation to teacher mobility.

Thank You & Contact Information

Thank you for your attention!

Chanh Bao Lam, Ph.D.
Kansas State University
cblam@k-state.edu